

Zhizheng Wang

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EDUCATION

University of Michigan - Ann Arbor, MI
M.S. in Data Science

Aug 2025 - May 2027(Expected)
GPA: 4.00/4.00

University of Georgia - Athens, GA
B.S. in Data Science and Cognitive Science | Minor: Computer Science
Magna Cum Laude, Classic City Scholar (4 years), Dean's List (8 Semesters)

Aug 2021 - May 2025
GPA: 3.85/4.00

SKILLS

Programming: Python (4 yrs), SQL (3 yrs), R (4 yrs), Java (4 yrs)

Tools: NumPy, Pandas, TensorFlow, PyTorch, Jupyter, Git/GitHub, Linux, LaTeX, Excel, Power BI

Skills: Machine Learning, Deep Learning, Computer Vision, Image Analysis, Data Visualization, Data Manipulation, Predictive Modeling, Reinforcement Learning, Natural Language Processing

Relevant Courseworks: Data Science I & II, Data Mining, Statistical Inference, Data Structures, Database Management, Data Security & Privacy, Artificial Intelligence, Natural Language Processing, Biomedical AI, Database Application Design, Regression, Time Series, & Categorical Data Analysis

WORK EXPERIENCE

Michigan Center for Youth Justice (MCYJ)

September 2025 - Present

Data Analyst

Ann Arbor, MI

- Developed an LLM-powered chatbot to provide public access to case information and policy documents
- Extracted features from juvenile justice case reports and analyzed trends in incidents and outcomes
- Built dashboards to visualize metrics, supporting data-driven, equitable youth justice decisions

Intelligent Thought and Action Lab (THINC) <http://thinc.cs.uga.edu>

February 2024 - September 2024

Undergraduate Student Researcher

Athens, GA

- Curated a 6,000+ image dataset, enabling a YOLO model to achieve 99% object detection accuracy
- Automated data workflows with 10+ Python scripts for camera calibration and data extraction
- Integrated and fine-tuned YOLOv8 into the PrecisionSort software to drive continuous performance gains

Office for Student Success and Achievement (OSSA)

January 2024 - May 2025

Peer Tutor (Statistics/Computer Science/Mathematics)

Athens, GA

- Provided constructive feedback and guided problem-solving in one-on-one and group Study Pod sessions
- Mentored over 50 students in academic strategies, enhancing critical thinking and analytical skills

Data Science & Artificial Intelligence Camp (Department of Statistics)

July 2024

Student Assistant

Athens, GA

- Guided 30 students in data science and coding, supporting instructors and fostering collaboration

PROJECTS

Addressing Data Imbalance: Upsampling vs. Downsampling on Sport Images

(Python)

- Resampled a 6,000+ image Sports dataset using NNI, Bilinear, Bicubic, Gaussian Pyramid, SRCNN, and Wavelet
- Improved complex deep learning models to 94% validation accuracy (GoogLeNet/ResNet50) and significantly boosted simpler CNNs, increasing AlexNet by 34% and VGG16 by 22%
- Generated synthetic high-res images with SRCNN/Wavelet to balance classes and improve feature retention

Urban Heat Island (UHI) Prediction Project

(Python)

- Predicted UHI intensity for 11,229 NYC locations using satellite imagery, building footprints, and weather data
- Built spatial features and analyzed data, identifying B01, NDVI, and building proximity as key predictors
- Modeled spatial patterns and neighborhood heat using ML, with Random Forest as the top model ($R^2 = 0.636$)

Stock Market Time Series Forecasting

(Python, R)

- Analyzed 40+ years (4,700+ trading days) of S&P 500 prices using R (quantmod, rugarch, fGarch, ggplot2)
- Tested 10+ ARMA-GARCH models, selecting ARMA(1,1)-GARCH(2,2) with AIC -6.53, improving fit by 12%
- Built and validated a 14-day volatility forecast on 2024 data; recommended log-transformation to boost accuracy

F-1 Racing

(HTML, CSS, JavaScript, SQL)

- Built and normalized SQL database for 73 years of F1 data (1950-2023) with 12+ tables and key constraints.
- Developed React frontend with 3 analytical queries and custom SQL input, handling 1M+ rows.
- Uncovered F1 trends: nationality dominance, circuit effects on lap times/pit stops, and top drivers on long tracks.

INTEREST

Interests: Weightlifting, Rock Climbing, Photography, Go, Reading, Attending Academic Seminars & Research Talks